FINITE LIFTS AND SIZE RESTRICTIONS FOR WEAK CENTRAL GROUPOIDS

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Abstract. The Equational Theories Project asks whether the order of every finite weak central groupoid is a square or twice a square. We characterize the possible fibre sizes of a complete lift of a complete marked base by positive integer factors on its edges. This proves the size restriction for every model whose bad two-step paths are detected by homomorphisms to the two-element NAND magma. A certified enumeration gives the same restriction for lifts of relaxed marked bases with at most five types. One five-type base has exact spectrum $\{16s^2 : s \geq 1\}$. At order twenty we rule out transitive automorphism groups by checking the 47 minimally transitive actions of that degree. We also give two obstructions to finite constructions, with Lean verification of the underlying arguments. Existence at order twenty in general remains open.

1. Weak central groupoids and complete lifts

A magma is a set with a binary operation. We write its product by juxtaposition. A weak central groupoid is a nonempty magma satisfying

$$x = (yx)\big(x(zy)\big) \qquad \text{for all } x, y, z. \tag{1}$$

The Equational Theories Project (ETP) numbers this identity E1485 and asks whether every finite model has order s^2 or $2s^2$ for a positive integer s [1, Section 14, question 7]. Both families occur. On $S \times S$ the product $(a, b)(c, d) = (b, c)$ satisfies the central groupoid identity $(xy)(yz) = y$, hence (1). Taking its direct product with the two-element NAND magma gives order $2|S|^2$ [2, Chapter 12]. Twenty belongs to neither family. We investigate the question through finite lifts and automorphism groups.

For a weak central groupoid, the ETP blueprint associates a directed graph by placing an edge $a \to b$ when $b = ac$ for some c. Equivalently, $a = db$ for some d. The path $a \to ab \to b$ is designated *good*. Any other two-step path is called *bad*. The blueprint shows that the following properties hold [2, Lemmas 12.2–12.4]. We write abc for a two-step path with middle vertex b, allowing loops and repeated vertices.

(1) Between any ordered pair of vertices there is a good path.
(2) Each edge begins a good path and ends a good path.
(3) In any closed walk $a \to b \to c \to d \to e \to a$, if abc and cde are good, then bcd is good.

A nonempty directed graph with a chosen set of good paths satisfying these conditions is called a *relaxed marked base*. Its vertices will be called *types*. A lift has a type map from its vertices to the base. Its edges project to edges and a two-step path in the lift is good if and only if its type triple is good in the base. A lift is *complete* if it has exactly one good path between every ordered pair. The type map is surjective in all lift statements below. Each edge of a lift is required to begin and end a good path, as in the marked-graph description. The unique good middle defines the product in a complete lift. Indeed, for $w = zy$, the good paths $y \to yx \to x$

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and $x \to xw \to w$ and the edge $w \to y$ imply $(yx)(xw) = x$ by the closed-walk rule. The blueprint uses this construction to obtain infinite models [2, Proposition 12.5 and Theorem 12.6].

The edge factorization in Theorem 3.1 gives a criterion for prescribed fibre sizes, and Theorem 3.2 applies it to quotients by NAND homomorphisms. For relaxed bases we obtain the following finite classification result.

Theorem 1.1. *Every finite complete lift of a relaxed marked base on at most five types has order s^2 or $2s^2$ for some positive integer s.*

Theorem 1.2. *The five-type marking specified in Section 5 has a finite complete lift of order n if and only if $n = 16s^2$ for some positive integer s.*

Theorem 6.1 rules out transitive automorphism groups at order twenty. Section 7 gives two additional construction obstructions. The complete-lift results use the exact rule for path markings. A magma homomorphism in general may send a bad path to a good path.

2. Counting paths in matrix blocks

Let n_i be the size of type i in a finite complete lift. Let A_{ij} be its $n_i \times n_j$ adjacency block and write J_{ij} for the all-ones block of that shape. Completeness gives

$$\sum_{k:\, ikj \text{ good}} A_{ik}A_{kj} = J_{ij}. \tag{2}$$

We use ordinary matrix multiplication over nonnegative integers. The arguments hold for arbitrary nonnegative integer entries satisfying the same equations.

We will use the following observation repeatedly. If $XY = J$ and $YZ = J$ for three compatible blocks, then $X(YZ) = (XY)Z$ implies that every row sum of X equals every column sum of Z. Products by all-ones blocks give the indicated row and column sums. All dimensions in such comparisons are positive.

2.1. **All-good markings and separated defects.** If every two-step path is good, the full adjacency matrix A of a lift satisfies $A^2 = J$. The equality $AJ = JA$ makes every row and column sum equal to a common integer r. Taking row sums in $A^2 = J$ gives $n = r^2$. This is the classical matrix count for central groupoids [9, 10].

Lemma 2.1. *Suppose $A^2 = J + K$, where K is nonnegative. If a row of both AK and KA vanishes and K has a zero column, then $K = 0$ and the dimension of A is square. The transposed assertion also holds.*

Proof. Associativity gives $AJ + AK = JA + KA$. On the specified row, the column sums of A are therefore all equal to the sum r of that row. Each column sum of A^2 is r^2, so each column sum of K equals $r^2 - n$. The zero column forces this common sum to vanish. Nonnegativity gives $K = 0$. □

In a lift, K counts the bad two-step paths. Let R be the set of types that begin a bad path in the base, and C the set of types that end one. A type outside R with no outgoing edge into R supplies the zero row of AK and KA. A type outside C supplies the zero column of K. This gives a test on the base alone. The catalogue also applies the transposed test.

2.2. **Two groups and twice-square order.** Suppose the types can be grouped into two nonempty parts L, H. Assume there are no edges within H and that a two-step path is good exactly when its vertices do not all lie in L. In the lift write

$$A = \begin{pmatrix} B & C \\ D & 0 \end{pmatrix}.$$

The path equations are

$$CD = J_{LL}, \qquad DC = J_{HH}, \qquad BC = J_{LH}, \qquad DB = J_{HL}. \tag{3}$$

This block formulation was discussed by Tao and Brox in the ETP collaboration [3]. We include the row-sum argument used in our tests. Reassociating $C(DC)$ and $D(CD)$ identifies each of the row and column sums of C, D with a constant for its block. The identity $D(BC) = (DB)C$ equates the two constants. Write their common value as r. Summing the rows of CD and DC gives $|L| = |H| = r^2$. In addition, $B(CD) = (BC)D$ makes every row sum of B equal to r, and $C(DB) = (CD)B$ makes every column sum of B equal to r.

2.3. **A forbidden three-group support.**

Lemma 2.2. *If $A^2 = J$, then A cannot have the block form*

$$A = \begin{pmatrix} B & C & 0 \\ D & 0 & E \\ 0 & F & G \end{pmatrix}$$

with all three parts nonempty.

Proof. Write ℓ, m, h for the three part sizes. The off-diagonal equations of $A^2 = J$ make BC, DB, CE, EG, FD and GF all-ones blocks. Comparing $D(BC)$, $E(GF)$, $C(EG)$ and $F(DB)$ with their reassociations gives the following constants.

$$\begin{aligned} \operatorname{rows}(D) = \operatorname{cols}(C) = d \quad & \operatorname{rows}(E) = \operatorname{cols}(F) = e \\ \operatorname{rows}(C) = \operatorname{cols}(G) = c \quad & \operatorname{rows}(F) = \operatorname{cols}(B) = b. \end{aligned}$$

Here each equality means that all the indicated sums equal the scalar on its right. Since A has constant row sums, B and G have constant row sums as well. Counting the entries of each square block by rows and columns shows that its row constant equals its column constant. Thus the row sums of B and G are b, c, respectively. Taking row sums in the first and third diagonal block equations, and in $DB = J$, $CE = J$, gives

$$b^2 + cd = \ell = db, \qquad be + c^2 = h = ce.$$

Positivity of ℓ, h gives $b, c > 0$. The first equation requires $b > c$, and the second requires $c > b$. □

2.4. **Saturation of a two-step count.** A good type triple ikj is called *sole* if it is the only good type triple from i to j. It gives $A_{ik}A_{kj} = J_{ij}$. In particular, every row of A_{ik} is nonzero. A pair of overlapping sole triples ikj, kjt identifies all row sums of A_{ik} with all column sums of A_{jt} by the observation above. We retain the equalities generated by these comparisons.

Suppose two blocks in an allowed term $A_{ik}A_{kj}$ have constant row sums r, s. If a sole term with target type j has the same pair of row constants, in either order, then $rs = n_j$. Every row of $A_{ik}A_{kj}$ therefore already has the row total required by (2). If another allowed term $A_{ib}A_{bj}$ has both factors with every row nonzero, it contributes a positive row total. This contradicts (2). The catalogue detects the latter condition when each factor begins a sole triple.

2.5. **Four groups with equal square sizes.** Partition the types into four nonempty groups indexed $0, 1, 2, 3$. For each indicated source and target group in the following array, require that an existing two-step path is good exactly when its middle lies in the displayed group. Dashes leave the path status free.

(F)

	0	1	2	3
0	3	–	–	0
1	2	2	–	0
2	1	–	1	0
3	0	–	–	0

For example, from group 1 to group 0 this condition gives $A_{12}A_{20} = J_{10}$, where the blocks are now indexed by groups. The entries for $0, 3$ give (3) with $B = A_{00}$, $C = A_{03}$, $D = A_{30}$. Thus $|0| = |3| = r^2$ for a positive integer r, and the row sums of C and the column sums of D are r. For $i = 1, 2$, compare $A_{i0}(CD)$ with $(A_{i0}C)D$. The equations $CD = J_{00}$ and $A_{i0}C = J_{i3}$ make every row sum of A_{i0} equal to r. The equations

$$A_{12}A_{20} = J_{10}, \qquad A_{21}A_{10} = J_{20}$$

then make the row sums of A_{12} and A_{21} equal to r, by cancellation of r in the row totals. Finally $A_{12}A_{21} = J_{11}$ and $A_{21}A_{12} = J_{22}$ give $|1| = |2| = r^2$. Thus the total order is $4r^2$.

3. An exact criterion for lifts of complete bases

A base is *complete* when each ordered pair has just one good middle. The complete-base hypothesis in this section concerns the base itself. A complete lift can also exist over a base that has several good middles for some pair, as in Section 5.

Theorem 3.1. *Let B be a finite complete marked base with vertex set V and let n_v be a positive integer for each $v \in V$. There is a complete lift with these fibre sizes if and only if one can assign positive integers q_{ab} to the directed edges of B so that*

$$q_{am}q_{mb} = n_m \qquad \text{for every good path } a \to m \to b. \tag{4}$$

Proof. Write A_e for the adjacency block associated to an edge e. If e, f form a good path, then $A_eA_f = J$ by (2). When e, f and f, g are both good, the row sums of A_e equal the column sums of A_g by associativity. For an edge e, choose a good successor f. Every edge begins and ends a good path, so there is also a good predecessor d of e. The row sums of A_d equal the column sums of A_f. The same holds for any other good successor of e. Thus these column sums are constant and their common value q_e is independent of the choice of successor.

For a good path $a \to m \to b$ with edges e, f, the row sums of A_e are q_f and the column sums of A_f are q_e. Let r be the common row sum of A_f, which exists because f has two successive good continuations. Taking row sums in $A_eA_f = J_{ab}$ and counting the entries of A_f in two ways give

$$q_f r = n_b, \qquad n_m r = n_b q_e.$$

Since $n_b > 0$, cancellation gives $n_m = q_e q_f$. Every edge begins a good path, so all factors are positive.

Conversely, label each fibre a by $0, \dots, n_a - 1$. For each base edge $a \to b$, the integer q_{ab} divides both endpoint sizes by (4) and the good-path support of the edge. Put an edge from (a, i) to (b, j) exactly when the base edge exists and

$$\left\lfloor \frac{i}{n_a/q_{ab}} \right\rfloor = j \bmod q_{ab}.$$

Mark paths by their projected type triples. For given endpoints $(a, i), (b, j)$, let m be their unique good middle type. By (4), the maps $k \mapsto k \bmod q_{am}$ and $k \mapsto \lfloor k/q_{am} \rfloor$ jointly identify a unique index k in fibre m with the required incoming and outgoing labels. Both endpoint label maps are surjective. Thus an edge extends on each side to a good path by choosing a good continuation of its base edge and then a vertex with the required label. The five-step rule pulls back from the base, so this is a complete lift. □

3.1. **When NAND maps detect every bad path.** Write N for the magma on $\{0, 1\}$ with product $a \star b = 1 - ab$ in ordinary bit arithmetic. The only missing edge is $0 \to 0$, and the only bad path is $1 \to 1 \to 1$. Say that a bad path of a weak central groupoid is *detected by NAND* if some magma homomorphism to N sends all three vertices to 1.

Theorem 3.2. *Let M be a finite weak central groupoid in which every bad path is detected by NAND. There are integers $k \geq 0$ and $r \geq 1$ with $|M| = 2^k r^2$. In particular its order is a square or twice a square.*

Proof. Combine all homomorphisms to N into one map to a product of copies of N. If there are no such homomorphisms, use the one-element product. The image is a finite Boolean subalgebra of the Boolean power because complement and intersection can be expressed using NAND. It is therefore isomorphic, with its NAND operation, to N^k for some $k \geq 0$. The map onto this image projects graph edges to edges and good paths to good paths. A bad path projects to a bad path because at least one coordinate detects it. Hence M is a complete lift of N^k.

Let u be the all-ones element of N^k and let $Sv = vv$, the bitwise complement. Every vertex has edges to and from u. For each edge $a \to b$, the paths $a \to b \to Sb$ and $Sa \to a \to b$ are good. Indeed $a \to b$ means that no coordinate of a, b consists of two zeros, which gives $a(Sb) = b$ and $(Sa)b = a$ coordinatewise. By (4) all edges entering b have the same factor, since they share the factor $q_{b,Sb}$. Similarly all edges leaving a have the same factor. Using u shows that every edge factor equals $q_{uu} = r$. Thus each of the 2^k fibres has size r^2. □

The hypothesis is automatic for central groupoids, whose paths are all good. It also holds in the extensions of NAND with all bad paths over $1 \to 1 \to 1$ discussed by Tao and Brox [3]. By contrast, a bad path containing x and xx cannot be detected by NAND. The ETP's twisted 32-element example fails $(xx)(xx) = x$ [2, Section 12.1]. At a failing x, the path $xx \to x \to xx$ is bad, and NAND sends x and xx to opposite bits. The theorem therefore gives a restriction on any counterexample to the size conjecture.

4. The finite catalogue

We enumerate marked bases on $\{0, \ldots, t-1\}$ for $1 \leq t \leq 5$. There is one propositional variable e_{ab} for each possible edge and g_{abc} for each possible good path. The constraints are

$$g_{abc} \Rightarrow e_{ab} \wedge e_{bc}, \qquad \bigvee_b g_{abc} \quad \text{for each } (a, c),$$

$$e_{ab} \Rightarrow \bigvee_c g_{abc}, \qquad e_{ab} \Rightarrow \bigvee_c g_{cab},$$

$$g_{abc} \wedge g_{cde} \wedge e_{ea} \Rightarrow g_{bcd}.$$

They encode exactly the relaxed marking axioms. Loops and repeated indices are included. For $t = 5$ we require the adjacency bit string to be lexicographically least among its vertex relabelings. For each permutation, auxiliary variables record agreement of the two strings up to a position, and a clause prohibits a first difference with the original bit larger. Every marking has a representative with this adjacency string. We then remove any remaining duplicates by taking the least pair of edge and good-path lists under simultaneous relabeling.

Each enumerated assignment is checked directly against the axioms. The blocked formula consists of the initial constraints and one clause excluding each recorded assignment to the edge and good-path variables. CaDiCaL 1.9.5 through PySAT produced a DRAT certificate of unsatisfiability for each blocked formula. We checked each certificate with DRAT-trim [11]. For five types, 23,689 assignments with least adjacency strings give 23,525 distinct markings after relabeling. The blocked-formula checks certify the completeness of the enumeration, and the ancillary verifier checks the association between the assignments, the catalogue, and the matrix tests.

For each marking the classification procedure is as follows.

(1) Check whether all paths are good or Lemma 2.1 forces $K = 0$. In that event, search for the forbidden support of Lemma 2.2.
(2) Test the saturation condition using the row and column equalities from the sole triples.

(3) Record the square restriction from the first step if it applies and neither obstruction was found.
(4) Search for a two-group partition satisfying (3) by the marking rule of Section 2.2, or a four-group partition satisfying (F).
(5) Coalesce types with identical behavior in every position of the edge and good-path predicates, and apply the same tests to the reduced marking.

In the last step, the predicates descend to the coarser type map because replacing any one coordinate preserves the predicate. The original complete lift is then a complete lift of the reduced marking. Each test only uses the marked base, so a successful classification applies to every finite lift using all of its types.

Types	Markings	Square required	Twice-square required	No finite lift
1	1	1	0	0
2	3	2	1	0
3	24	16	6	2
4	388	296	71	21
5	23,525	20,424	2,669	432

TABLE 1. Marked bases up to relabeling, assigned to the first applicable conclusion in the classification procedure. A required size form is a necessary condition for a lift.

Table 1 reports all outcomes. Every marked base was classified, which proves Theorem 1.1. The same tests exclude all four valid markings on the particular five-vertex graph in the proof of the ETP blueprint's Theorem 12.6, including the marking used there for an infinite construction.

5. A base with exact spectrum $16s^2$

Use types $0, 1, 2, 3, 4$, with edges in both directions for each unordered pair

$$01,\ 02,\ 03,\ 04,\ 12,\ 13,\ 14,\ 23.$$

There are no loops. The sets of good middles are the entries in this array, indexed by source and target.

(G)

	0	1	2	3	4
0	$\{4\}$	$\{4\}$	$\{3\}$	$\{2\}$	$\{1\}$
1	$\{4\}$	$\{4\}$	$\{3\}$	$\{2\}$	$\{0\}$
2	$\{3\}$	$\{3\}$	$\{3\}$	$\{0,1\}$	$\{0,1\}$
3	$\{2\}$	$\{2\}$	$\{0,1\}$	$\{2\}$	$\{0,1\}$
4	$\{1\}$	$\{0\}$	$\{0,1\}$	$\{0,1\}$	$\{0,1\}$

The finite marking check from Section 4 verifies the relaxed axioms for these data.

Group the types as $\{0,1\}, \{2\}, \{3\}, \{4\}$ in that order. The coarser marking is the graph of N^2, with the groups labeled $(1,1), (1,0), (0,1), (0,0)$. It satisfies (F), so each group has size r^2. Moreover the internal block B of the first group is supported across the partition into types 0 and 1. Its row and column sums are r by (3) and the ensuing calculation. Counting entries in the block from type 0 to type 1 by rows and columns shows $n_0 r = n_1 r$, hence $n_0 = n_1$. Since $n_0 + n_1 = r^2$, the integer r is even. Writing $r = 2s$ gives total order $4r^2 = 16s^2$.

Appendix A gives a 16-element complete lift with type sizes $(2, 2, 4, 4, 4)$. Its table satisfies (1), and its graph has exactly the good paths prescribed by (G). Both claims are checked directly by the ancillary witness verifier. To obtain every larger allowed order, take the direct product of

this lift with the natural central groupoid on $S \times S$, where $|S| = s$. In a direct product the graph edges are componentwise, and every two-step path in the central groupoid is its unique good path between those endpoints. A path in the product is therefore good exactly when its path in the 16-element factor is good. The product is a complete lift with the same base and has order $16s^2$. This proves Theorem 1.2.

6. The transitive case at order twenty

An automorphism of a magma is a permutation preserving its product. The automorphism group is transitive if it can send any element to any other. This condition is weaker than the existence of a cyclic permutation of length twenty preserving the product.

Theorem 6.1. *No weak central groupoid of order twenty has a transitive automorphism group.*

Proof. A finite transitive permutation group contains a subgroup that is minimal among its transitive subgroups. Conjugating such a subgroup amounts to relabeling the elements of the magma. Hulpke's classification lists 47 minimally transitive actions of degree twenty up to conjugacy [6, 7, 8]. There are five regular and 42 nonregular actions.

The regular actions are the left translations of groups of order twenty. In each such group the Sylow 5-subgroup is normal, and a Sylow 2-subgroup of order four is a complement. For a cyclic complement the action on the normal subgroup can have image of order $1, 2, 4$ in $\mathrm{Aut}(C_5)$, with the two possible generators of order four related by a change of generator of the complement. For a Klein-four complement there is a trivial action and one nontrivial action up to changes of basis. These are the five regular cases. For the nonregular cases we use the generators in Hulpke's degree-twenty table [7].

Fix one of these actions G on $\Omega = \{0, \dots, 19\}$. Use one Boolean literal $T(x, y, z)$ per orbit of ordered triples to express $xy = z$. For each x, y the list $(T(x, y, z))_{z \in \Omega}$ has exactly one true entry. The list is indexed by output elements, so a literal appearing twice is forbidden by the at-most-one clauses. This describes all G-invariant operation tables. Define a Boolean literal $C(w, y)$, indexed by ordered pair orbits, to be true exactly when some z has $T(z, y, w)$ true. Auxiliary triple-orbit literals $R(x, y, b)$ are subject to

$$C(w, y) \wedge T(x, w, b) \implies R(x, y, b),$$
$$T(y, x, a) \wedge R(x, y, b) \implies T(a, b, x).$$

Any satisfying assignment gives (1) by taking $w = zy$. Conversely, a table satisfying (1) extends to such an assignment by making $R(x, y, b)$ true precisely when there is a w in the column at y with $xw = b$. All clauses are equivariant. It suffices to set $x = 0$ in the two implications and to emit the functionality and column clauses with their first index fixed to 0.

Every one of the 47 formulas was unsatisfiable. Each clausal proof was checked by DRAT-trim [11]. The ancillary verifier reconstructs the formulas from the generators and checks the supplied proof traces. It follows that none of the minimally transitive actions preserves a table of order twenty satisfying (1). □

Together, Theorems 1.1, 3.2, and 6.1 give three necessary conditions for a model of order twenty. A complete marked-base representation must use at least six types, the model must have a bad path undetected by all NAND maps, and its automorphism action must have at least two orbits. The existence of an unrestricted model of order twenty remains open.

7. Obstructions to two constructions

7.1. **A finite obstruction for Bolan's marking.** Bolan posted a relaxed marking with seventeen vertices containing a bad path of types compatible with failure of E166 [5]. In

any finite complete lift, group the types into a core $K = \{0, 1, 2, 13, 15\}$, a leaf group $L = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 16\}$, and a hub $H = \{14\}$. Write the aggregated adjacency matrix in the form

$$\begin{pmatrix} B & C & 0 \\ D & 0 & E \\ 0 & F & G \end{pmatrix}.$$

For the ordered group pairs listed below, a two-step path is good exactly when its middle lies in one of the displayed groups:

(source, target)	(K,H)	(L,K)	(L,L)	(L,H)	(H,L)	(H,H)
allowed middle	L	K	K, H	H	H	L

In addition, for every individual leaf type the hub is the only good middle type from that type to itself. These properties are checked directly against the posted marking in the ancillary verifier.

Proposition 7.1. *Bolan's seventeen-type marking has no finite complete lift.*

Proof. The marked paths give $EG = J$, $GF = J$, $FE = J$, and $DC + EF = J$, with the appropriate block shapes. Each diagonal entry of EF equals one by the leaf type condition. The equations $DB = J$ and $CE = J$ make every row of D and C positive in total.

Comparing $E(GF)$ with $(EG)F$ makes all row sums of E equal to a constant e. Comparing $F(EG)$ with $(FE)G$ does the same for the row sums of F, with constant f. Thus each row sum of EF is ef, which by $FE = J$ equals the size of the hub group. The traces of EF and FE agree by interchanging their two summation indices. Their all-ones diagonals then give equal leaf and hub sizes. It follows that EF supplies the entire row sum required by $DC + EF = J$, whereas each row sum of DC is positive. This contradiction excludes the lift. □

7.2. **Removing the diagonal from a natural central groupoid.** The natural central groupoid on $S \times S$ has product $(a, b)(c, d) = (b, c)$. When $|S| = 5$, removing the diagonal leaves twenty elements. One could seek to preserve the old product wherever it still takes values off the diagonal.

Proposition 7.2. *Let $|S| \geq 3$ and put $X = \{(a, b) \in S^2 : a \neq b\}$. There is no weak central groupoid on X whose product equals (b, c) whenever the inputs are $(a, b), (c, d)$ with $b \neq c$.*

Proof. Write $x = (a, b)$ and $\bar{x} = (b, a)$. The retained products give $xx = \bar{x}$ and $\bar{x}\bar{x} = x$. Put $m = \bar{x}x$. Using (1) with inputs $(x, \bar{x}, \bar{x})$ and $(\bar{x}, \bar{x}, \bar{x})$ gives $m\bar{x} = x$ and $xm = \bar{x}$, respectively. If the second coordinate of m differs from b, the first equality forces it to equal a by the retained rule. If the first coordinate of m differs from b, the second equality forces it to equal a by the same rule. Since m lies off the diagonal, it is x or $\bar{x}$.

If $\bar{x}x = x$, then for every $y = (b, f)$ the retained rule gives $x(y\bar{x}) = y$. Applying (1) with inputs $(x, \bar{x}, y)$ gives $xy = x$, including $x\bar{x} = x$. If $x\bar{x} = x$, then for every $y = (c, a)$ the retained rule gives $x\bar{y} = \bar{x}$ and $\bar{y}x = y$. Applying the identity with inputs $(x, \bar{y}, x)$ gives $yx = x$, including $\bar{x}x = x$.

For each unordered pair $\{a, b\}$, orient the pair in the direction of the element to which both products $x\bar{x}$ and $\bar{x}x$ are equal. This is well defined by the preceding argument applied to x or $\bar{x}$. Two consecutive oriented edges $(a, b), (b, c)$ would each force their product to equal itself, by the two consequences just proved. They are distinct, a contradiction. Every orientation of the three edges on three distinct labels contains a directed path of length two, which proves the assertion. □

8. Verification and scope

The ancillary material includes the enumeration data, multiplication table, group generators, proof traces, and verification code. CaDiCaL 1.9.5 via PySAT was used to produce the clausal proofs, and DRAT-trim checks them. The catalogue verifier reconstructs the blocked formulas and checks the classification data. The transitivity verifier reconstructs all 47 formulas and checks the traces. That theorem uses Hulpke's classification as an external input.

The Lean 4.35.0 sources check the matrix arguments for the catalogue and the parity restriction, necessity of the edge factorization, the lift product and exact marking, and the uniform square fibres over NAND powers. They also check the matrix obstruction used for Bolan's marking and the diagonal-removal argument. The finite Boolean image argument is proved in the text. The exhaustive enumerations, their clausal certificates, and the checks of specific finite tables and markings are separate from the Lean proofs.

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Appendix A. The 16-element lift

The elements are $0, \dots, 15$, in row and column order. The types in order are the intervals $\{0,1\}$, $\{2,3\}$, $\{4,5,6,7\}$, $\{8,9,10,11\}$, and $\{12,13,14,15\}$. An entry in row x, column y is xy.

$\times$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	12	13	13	12	9	9	8	8	7	6	6	7	2	2	3	3
1	15	14	14	15	9	9	8	8	4	5	5	4	2	2	3	3
2	12	13	13	12	11	11	10	10	7	6	6	7	0	0	1	1
3	15	14	14	15	11	11	10	10	4	5	5	4	0	0	1	1
4	8	8	11	11	11	11	8	8	1	1	2	2	2	2	1	1
5	9	9	10	10	9	9	10	10	0	0	3	3	0	0	3	3
6	9	9	10	10	9	9	10	10	0	0	3	3	0	0	3	3
7	8	8	11	11	11	11	8	8	1	1	2	2	2	2	1	1
8	6	7	7	6	1	1	0	0	7	6	6	7	0	0	1	1
9	5	4	4	5	1	1	0	0	4	5	5	4	0	0	1	1
10	6	7	7	6	3	3	2	2	7	6	6	7	2	2	3	3
11	5	4	4	5	3	3	2	2	4	5	5	4	2	2	3	3
12	3	3	0	0	3	3	0	0	0	0	3	3	0	0	3	3
13	2	2	1	1	1	1	2	2	1	1	2	2	2	2	1	1
14	2	2	1	1	1	1	2	2	1	1	2	2	2	2	1	1
15	3	3	0	0	3	3	0	0	0	0	3	3	0	0	3	3

References

[1] M. Bolan et al., *The Equational Theories Project: Advancing Collaborative Mathematical Research at Scale*, arXiv:2512.07087v2 (2025).

[2] Equational Theories Project contributors, *Equational Theories Project blueprint*, Chapter 12, Weak central groupoids.

[3] Equational Theories Project contributors, *Discussion of E1485*, Lean Zulip archive, discussion on 5 November 2024.

[4] T. Tao and J. Brox, *Order-sixteen examples in the discussion of E1485*, Lean Zulip archive, 24 October 2024 and 25 October 2024.

[5] M. Bolan, *A seventeen-vertex relaxed marking for E1485*, Lean Zulip archive, 28 October 2024.

[6] A. Hulpke, *Constructing transitive permutation groups*, J. Symbolic Comput. **39** (2005), no. 1, 1–30. Publisher record.

[7] A. Hulpke, *The minimally transitive groups of degree up to 30*, companion data to [6].

[8] A. Hulpke and contributors, *TransGrp: Library of transitive groups*, GAP package, minimally transitive index table.

[9] D. E. Knuth, *Notes on central groupoids*, J. Combinatorial Theory **8** (1970), no. 4, 376–390. Publisher record.

[10] F. Curtis, J. Drew, C.-K. Li, and D. Pragel, *Central groupoids, central digraphs, and zero-one matrices A satisfying* $A^2 = J$, J. Combin. Theory Ser. A **105** (2004), no. 1, 35–50. Publisher record.

[11] N. Wetzler, M. J. H. Heule, and W. A. Hunt Jr., *DRAT-trim: Efficient checking and trimming using expressive clausal proofs*, Theory and Applications of Satisfiability Testing – SAT 2014, Lecture Notes in Comput. Sci., vol. 8561, Springer, 2014, pp. 422–429. Publisher record.